

# AI Supply Chain: Identifying the Key Beneficiaries of CPO

We believe the Rubin GPU and the CPO design of its server rack system will change the optical networking chip landscape. Near-term, Blackwell GPU demand is intact with changing mix of TSMC CoWoS.

## Key Takeaways

- Initiate FOCI at OW for its fiber array unit expertise and raise PT for ASE on CPO packaging opportunity.

**Co-packaged optics (CPO) to unleash data transfer for AI datacenter:** Broadcom, Marvell and NVIDIA (covered by Joe Moore) are the major spearheads of global CPO chip design. As NVIDIA's Rubin and its NVL server rack system have better visibility of CPO introduction and larger content per system (75% of global CPO demand in 2027e), our Asia supply chain stock ideas focus more on beneficiaries of NVIDIA's Rubin CPO.

Our base case assumes the CPO market grows from US\$8mn to US\$9.3bn, suggesting a 172% 2023-2030e CAGR. We initiate [FOCI at OW](#) given its unique position within the CPO supply chain and the significant volume opportunity coming from NVIDIA's Rubin. We also raise our ASE PT given its SiP opportunity for Rubin's server rack system and reiterate OW on TSMC, AllRing and Chroma as being key CPO supply chain players. Although we are EW on Landmark and VPEC, SiPh and CPO are also key growth drivers for them heading into 2027.

**1. FOCI supplies key components FAU and related services to fulfill significant CPO demand:** We expect FOCI to be the main source of FAU and related services (optical packaging, optical fiber-jump-in-systems and parch cords), which we expect to be introduced in NVIDIA's next gen Rubin GPU server rack system. Besides its leading technology, we think the ability to meet TSMC's localization strategy helps FOCI stand out among its peers, and we expect significant CPO-related revenue contribution over 2026/27.

**2. ASE (3711.TW) provides CPO solutions across traditional to advanced packaging; PT up to NT\$198:** For CPO, we think ASE/SPIL could offer services incl.: 1) Hybrid bonding for PIC/EIC leveraging its VIPack platform; 2) PIC/EIC/Switch/XPUs OSAT services; and 3) SiP packaging for entire CPO chips. Although Shun Sin (6451.TW; NC) provided SiP packaging for Broadcom's TH5 ([Exhibit 4](#)), we view ASE as being quite proactive and expect some potential CPO opportunities. In [our FOCI report](#), we indicate that we expect there to be around 650k and 2,275k units of the components (including CPU, GPU, switch IC, etc.) needing CPO in 2026 and 2027, respectively. We assume US\$20 each for ASE SiP packaging, generating around NT \$400mn and NT\$1.4bn for ASE in 2026 and 2027.

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| GREATER CHINA TECHNOLOGY SEMICONDUCTORS |            |            |
|-----------------------------------------|------------|------------|
| Asia Pacific                            |            |            |
| Industry View                           |            | In-Line    |
| WHAT'S CHANGED                          |            |            |
| ASE Technology Holding Co.              |            |            |
| Ltd. (3711.TW)                          | From       | To         |
| Price Target                            | NT\$180.00 | NT\$198.00 |

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**3. AllRing (6187.TWO, OW) – CPO could be next growth driver for AllRing into 2026:** Silicon photonics has become another area of focus for AllRing's R&D (as much as 10% of labor resource working on this), as the technology in addition to AOI requires coupling and algorithms. We do see AllRing having good progress in CPO-related equipment development, and we believe CPO could become a key business opportunity for AllRing into 2026 ([link](#)). We believe AllRing's FAU optical coupling equipment could start to see significant revenue contribution in 1H26, as we expect FAU clients to buy equipment at least 1-2 quarters ahead of mass production ([see AllRing note](#)).

**4. TSMC (2330.TW, OW) leveraging COUPE to integrate optical and semiconductor market:** TSMC debuted its universal COUPE (COmpact Universal Photonic Engine) technology in 2021, an advanced packaging solution targeted at silicon photonics (SiPh) applications to consolidate different requirements onto the same integration platform. Under COUPE technology, TSMC offers EIC/PIC fabrication and optical chip integration using its SoIC solution. We expect slight revenue contribution from this in 2026 with a much stronger contribution beyond. Broadcom's Tomahawk (TH) 5-Baily PIC and 3DIC are currently manufactured by Silterra (private) but we cannot rule out the possibility of the next generation being provided by TSMC.

**5. MediaTek (2454.TW, OW) and Alchip (3661.TW, OW) include CPO in its ASIC design service offering:** MediaTek launched its next-generation ASIC design platform with co-packaged optics solutions at OFC 2024, where it demonstrated a serviceable socketed implementation integrating both MediaTek's in-house SerDes for electrical I/O and co-packaged Odin optical engines from Ranovus (private) for optical I/O. According to the company, its CPO demonstration delivers reduced device costs, boosts bandwidth density and lowers system power by up to 50% vs. existing solutions. Alchip also provides 3D packaging design solutions; these are included in its 2nm ASIC platform solution portfolio and could fulfill long-tail CPO demand.

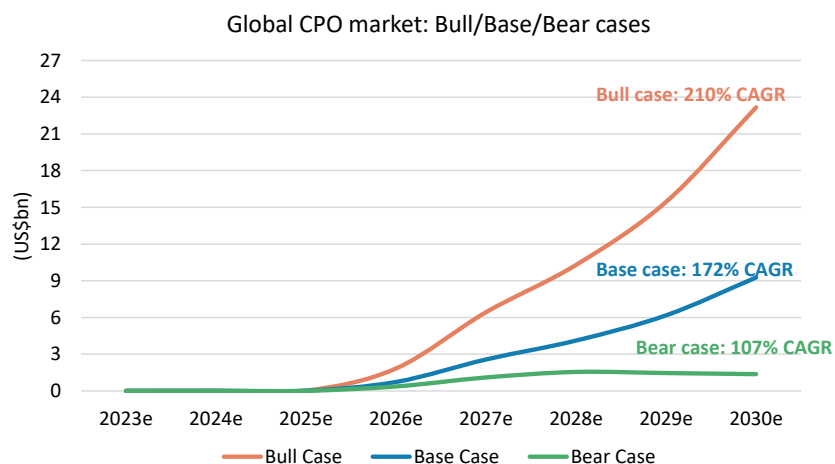
**6. Landmark (3081.TWO, EW) and VPEC (2455.TW, EW) provide SiPh epitaxial wafer work, which could be applied to CPO solutions:** LandMark has been offering silicon photonics (SiPh) epitaxial wafer work since 2012 and currently supplies SiPh epiwafer to large US ISP clients, focusing on high-performance computing (HPC) for AI-related development. The company sees SiPh structure as the most suitable solution for laser diodes in CPO design but expects CPO to be a long-term story, likely ramping up in 2027.

VPEC has been working with Japanese clients on SiPh projects for more than five years, with the project entering mass production in 4Q24. This business area started contributing revenue in 2024, and management expect it to grow in high multiples this year.

**7. Chroma Ate (2360.TW, OW) can provide photonics testing equipment for CPO:** Chroma has been supplying photonic testing equipment for laser diodes used in both 3D sensing and optical communications. Chroma can leverage this capability in the CPO configuration. We believe the revenue contribution should become more significant when CPO is more widely used.

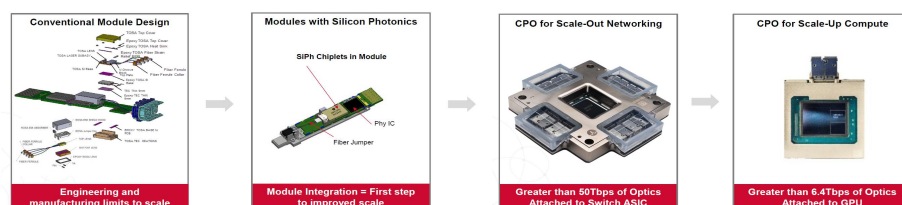
## Key charts for CPO developments

**Exhibit 1:** We expect CPO market revenue to grow rapidly, from US\$8mn to US\$9.3bn, implying 172% CAGR over 2023-2030e



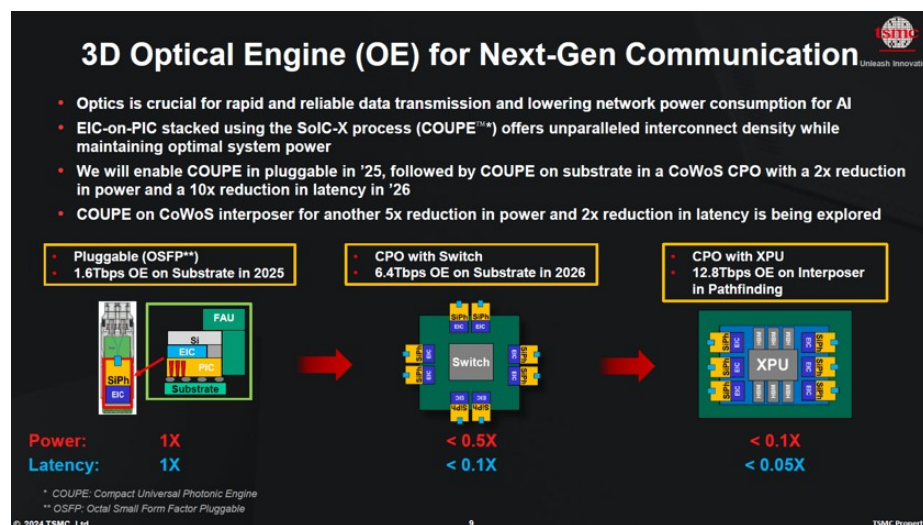
Source: Yole, Morgan Stanley Research (e) estimates

**Exhibit 2:** CPO helps to reach scale-out networking and scale-up computing goal for AI/HPC applications



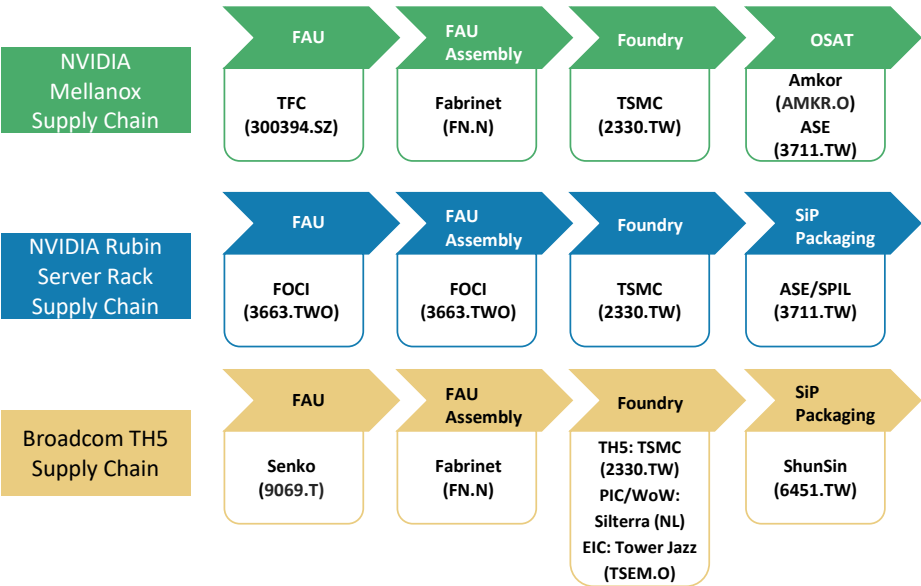
Source: Broadcom

**Exhibit 3:** TSMC's COUPE technology: Across SiPh pluggables to 3DIC



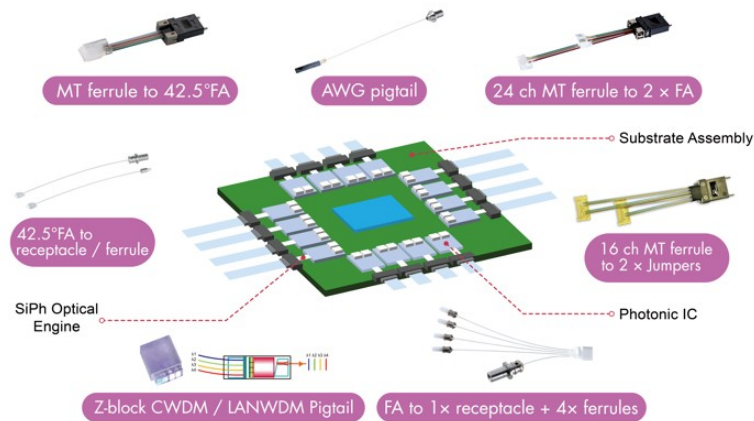
Source: TSMC

Exhibit 4: NVIDIA and Broadcom supply chain dynamics



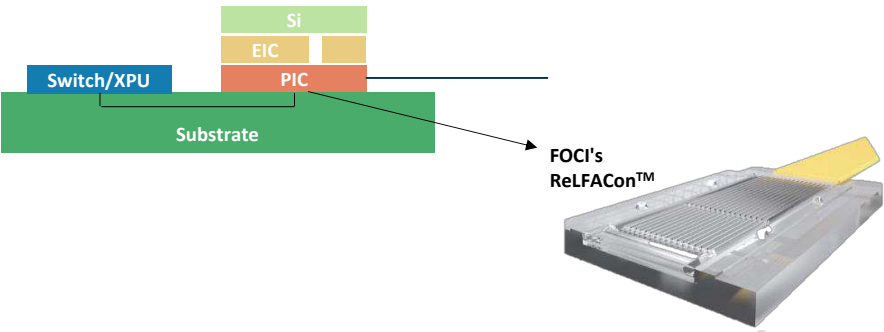
Source: FOCI, Morgan Stanley Research

Exhibit 5: FOCI's solutions for CPO



Source: FOCI

Exhibit 6: ReLFACon enables direct transmission of external photonic signals with MCM modules to achieve reliable signal transmission



Source: FOCI

# AI Semi Chip Supply Indicator – CoWoS Capacity Trend

## AI semi supply chain data point

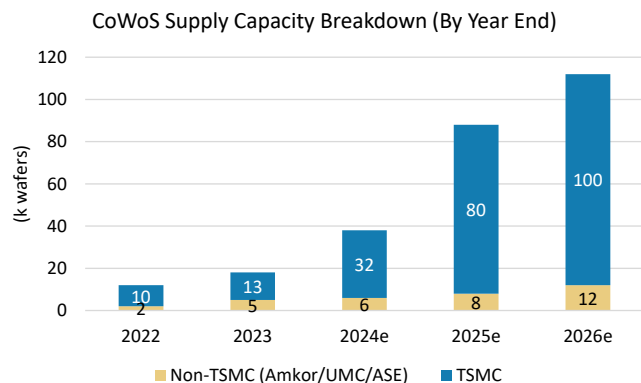
- Some of TSMC's customers such as AMD and Broadcom (both covered by Joe Moore) are releasing their CoWoS-S capacity (we think 10k-20k in total) for MI325 and Google TPU, respectively, given weak demand.
- NVIDIA has decided to take the released CoWoS-S, but has requested TSMC to convert it into CoWoS-L for better performance for its upcoming GB300A (the single GPU version) production. We estimate a total of around 60k of CoWoS-S will be converted to CoWoS-L. Going forward, all of NVIDIA's Blackwell GPUs will use CoWoS -L (including GB200, GB300, and GB300A).
- While TSMC's total CoWoS demand is unchanged, there should be slight upside for Nvidia's GB300A chip output in 2H25. The conversion from CoWoS-S to CoWoS-L will likely benefit equipment vendors like ASMPT, currently as a sole supplier for TSMC CoWoS-L oS (on substrate) TCB.
- Overall, cloud AI capex is still robust in 2025, but more skewed to NVIDIA's GPU purchase, echoing our ASIC 2.0 insight that NVIDIA GPU gains share from AMD GPU and Google's TPU this year.

**Exhibit 7:** Global CoWoS demand breakdown in 2025

| (k wafer)           | 2022      | 2023       | 2024e      | 2025e      | 2022e       | 2023e       | 2024e       | 2025e       |
|---------------------|-----------|------------|------------|------------|-------------|-------------|-------------|-------------|
| NVIDIA              | 24        | 53         | 200        | 465        | 40%         | 45%         | 54%         | 64%         |
| Broadcom            | 15        | 23         | 68         | 85         | 25%         | 20%         | 18%         | 12%         |
| AMD                 | 1         | 7          | 40         | 50         | 2%          | 6%          | 11%         | 7%          |
| Xilinx              | 2         | 3          | 10         | 10         | 3%          | 3%          | 3%          | 1%          |
| AWS + Alchip        | 0         | 9          | 16         | 18         | 1%          | 8%          | 4%          | 2%          |
| Intel Habana        | 0         | 0          | 7          | 12         | 0%          | 0%          | 2%          | 2%          |
| Marvell             | 2         | 1          | 18         | 70         | 3%          | 1%          | 5%          | 10%         |
| GUC                 | 0         | 1          | 1          | 1          | 0%          | 1%          | 0%          | 0%          |
| Others              | 16        | 20         | 10         | 19         | 27%         | 17%         | 3%          | 3%          |
| <b>Total demand</b> | <b>60</b> | <b>117</b> | <b>370</b> | <b>730</b> | <b>100%</b> | <b>100%</b> | <b>100%</b> | <b>100%</b> |

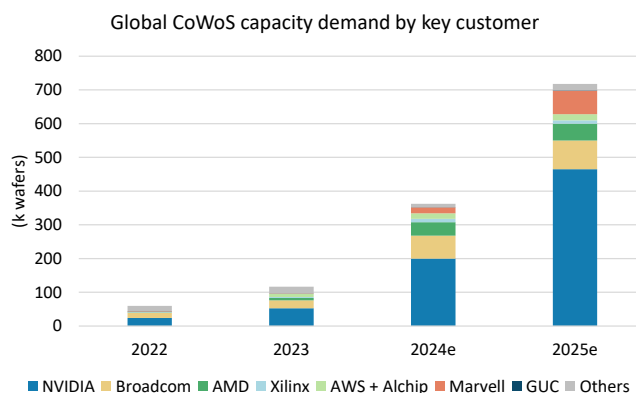
Source: Company data, Morgan Stanley Research (e) estimates

**Exhibit 8:** Global CoWoS capacity expansion by year-end and by vendor



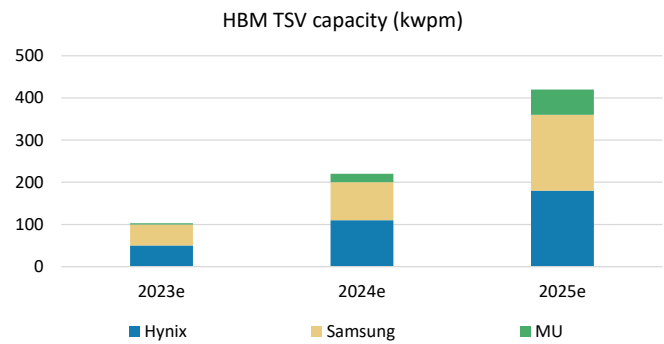
Source: Company data, Morgan Stanley Research (e) estimates

**Exhibit 9:** Global CoWoS consumption, by customer



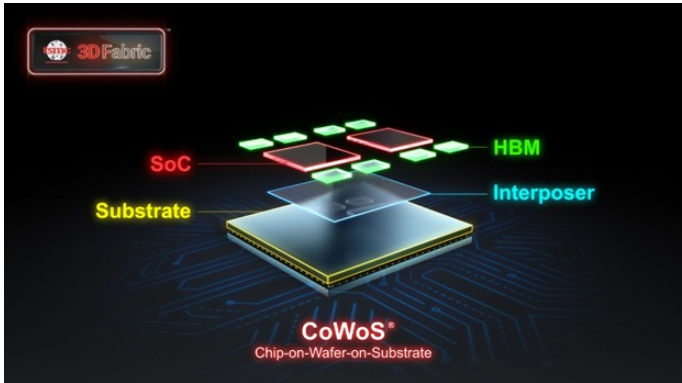
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 10: HBM TSV capacity appears set to double in 2025



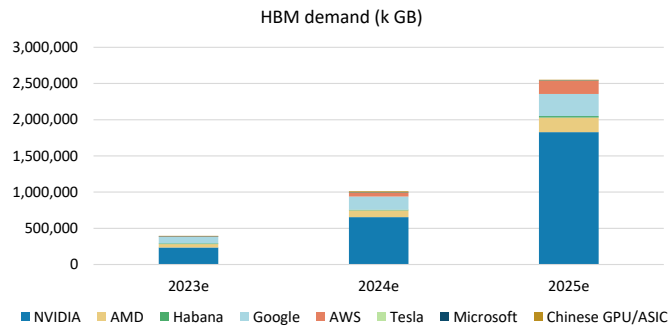
Source: TrendForce, Morgan Stanley Research (e) estimates

Exhibit 12: What is CoWoS? Chip on Wafer (interposer) on Substrate



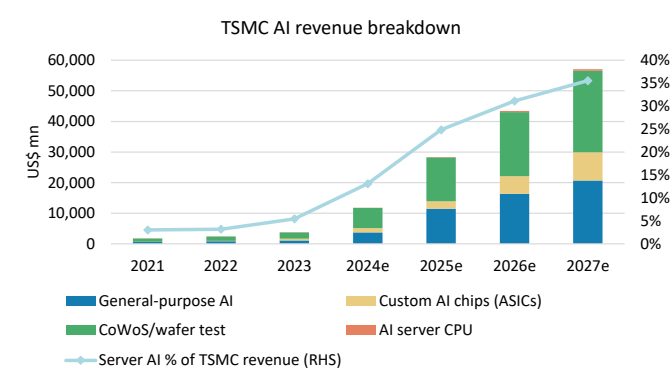
Source: TSMC

Exhibit 11: HBM demand is nearly twice the 2024 level in 2025



Source: TrendForce, Morgan Stanley Research (e) estimates

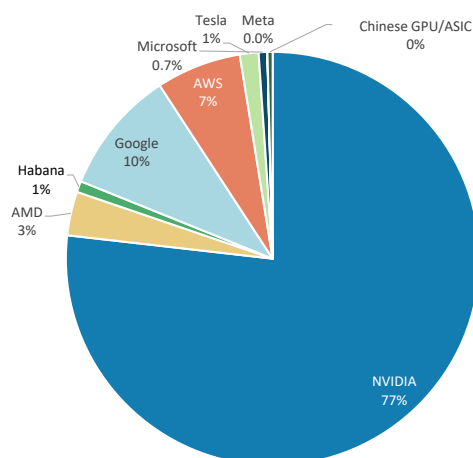
Exhibit 13: AI accounts for a mid-teens share of TSMC's 2024e total revenue and 25% of TSMC's 2025e total revenue



Source: Company data, Morgan Stanley Research (e) estimates

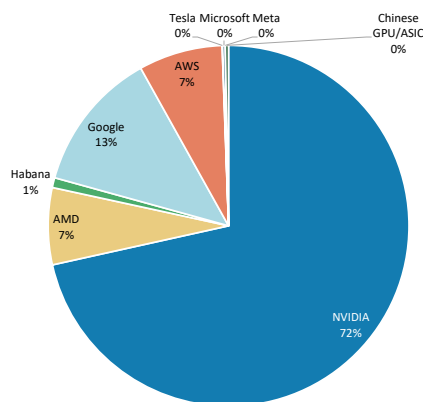
## 2025 AI semi foundry wafer revenue and HBM demand calculation

**Exhibit 14:** Leading-edge wafer consumption, by customer, 2025e



Source: Company data, Morgan Stanley Research (e) estimates

**Exhibit 15:** HBM consumption in 2025e – NVIDIA seen as the largest customer



Source: Company data, Morgan Stanley Research (e) estimates

**Exhibit 16:**

AI computing wafer consumption in 2025e – up to US\$14.6bn revenue

| AI chip vendor         | Product name         | CoWoS capacity allocation (k wafers) | Chips per CoWoS wafer | Implied shipments (k) | Compute die size | Geometry | Compute die units | Wafer consumption (k wafers) | Wafer price (US\$) | Wafer revenue TAM (US\$ mn) |
|------------------------|----------------------|--------------------------------------|-----------------------|-----------------------|------------------|----------|-------------------|------------------------------|--------------------|-----------------------------|
| <b>AI GPU (2025e)</b>  |                      |                                      |                       |                       |                  |          |                   |                              |                    |                             |
| NVIDIA                 | H100                 | 20                                   | 29                    | 580                   | 814              | 4nm      | 1                 | 16                           | 20,900             | 339                         |
|                        | H200                 | 100                                  | 29                    | 2,900                 | 814              | 4nm      | 1                 | 81                           | 20,900             | 1,695                       |
|                        | B200                 | 220                                  | 16                    | 3,520                 | 850              | 4nm      | 2                 | 279                          | 20,900             | 5,839                       |
|                        | B300                 | 125                                  | 16                    | 2,000                 | 850              | 4nm      | 2                 | 159                          | 20,900             | 3,317                       |
| AMD                    | MI300                | 5                                    | 12                    | 60                    | 110              | 5nm      | 8                 | 2                            | 25,200             | 44                          |
|                        | MI325                | 20                                   | 12                    | 240                   | 110              | 5nm      | 8                 | 7                            | 25,200             | 176                         |
|                        | MI355                | 25                                   | 12                    | 300                   | 110              | 3nm      | 8                 | 9                            | 32,004             | 279                         |
| Habana                 | Gaudi 3              | 12                                   | 14                    | 168                   | 750              | 5nm      | 2                 | 6                            | 20,000             | 126                         |
| <b>AI ASIC (2025e)</b> |                      |                                      |                       |                       |                  |          |                   |                              |                    |                             |
| Google                 | TPU v6               | 85                                   | 25                    | 2,125                 | 400              | 5nm      | 2                 | 64                           | 22,000             | 1,414                       |
|                        | Trainium 2           | 40                                   | 16                    | 640                   | 600              | 5nm      | 2                 | 24                           | 22,000             | 521                         |
| AWS                    | Trainium 2.5         | 30                                   | 16                    | 480                   | 600              | 5nm      | 2                 | 16                           | 22,000             | 361                         |
|                        | Trainium 3           | 18                                   | 10                    | 180                   | 700              | 3nm      | 1                 | 3                            | 27,940             | 86                          |
| Tesla                  | Dojo 1               | NA                                   | NA                    | 200                   | 645              | 7nm      | 1                 | 3                            | 15,400             | 54                          |
|                        | FSD                  | NA                                   | NA                    | 1,600                 | 300              | 7nm      | 1                 | 10                           | 15,400             | 160                         |
| Microsoft              | Maia 200             | 2                                    | 29                    | 58                    | 700              | 3nm      | 1                 | 1.5                          | 26,400             | 41                          |
|                        | Maia 200 Enhancement | 1                                    | 29                    | 29                    | 700              | 3nm      | 1                 | 0.8                          | 26,400             | 20                          |
| Chinese GPU/ASIC       | Chinese GPU/ASIC     | 5.5                                  | 20                    | 110                   | 1,000            | 7nm      | 1                 | 3                            | 18,700             | 59                          |
| <b>Total</b>           |                      | <b>709</b>                           |                       |                       |                  |          |                   | <b>688</b>                   |                    | <b>14,570</b>               |

Source: Company data, Morgan Stanley Research (e) estimates

**Exhibit 17:** HBM consumption in 2025e – up to 19.5bn Gb

| AI chip vendor                  | Product name         | CoWoS capacity allocation (k wafers) | Chips per CoWoS wafer | Implied shipments (k) | HBM chip density (GB) | HBM chip units | Total HBM size (GB) | HBM generation | HBM vendor           | Total HBM demand (k GB) |
|---------------------------------|----------------------|--------------------------------------|-----------------------|-----------------------|-----------------------|----------------|---------------------|----------------|----------------------|-------------------------|
| <b>AI GPU (2025e)</b>           |                      |                                      |                       |                       |                       |                |                     |                |                      |                         |
| <b>NVIDIA</b>                   | H100                 | 20                                   | 29                    | 580                   | 16                    | 5              | 80                  | HBM3           | Hynix                | 46,400                  |
|                                 | H200                 | 100                                  | 29                    | 2,900                 | 24                    | 6              | 141                 | HBM3e          | MU                   | 408,900                 |
|                                 | B200                 | 220                                  | 16                    | 3,520                 | 25                    | 8              | 200                 | HBM3e          | Hynix/Micron/Samsung | 704,000                 |
|                                 | B300                 | 125                                  | 16                    | 2,000                 | 36                    | 8              | 288                 | HBM3e 12hi     | Hynix/Micron/Samsung | 576,000                 |
| <b>AMD</b>                      | MI300                | 5                                    | 12                    | 60                    | 24                    | 8              | 192                 | HBM3           | Samsung              | 11,520                  |
|                                 | MI325                | 20                                   | 12                    | 240                   | 36                    | 8              | 288                 | HBM3e 12hi     | Samsung              | 69,120                  |
|                                 | MI355                | 25                                   | 12                    | 300                   | 36                    | 8              | 288                 | HBM3e 12hi     | Samsung              | 86,400                  |
| <b>Habana</b>                   | Gaudi 3              | 12                                   | 14                    | 168                   | 16                    | 8              | 128                 | HBM2e          | Samsung              | 21,504                  |
| <b>AI ASIC (2025e)</b>          |                      |                                      |                       |                       |                       |                |                     |                |                      |                         |
| <b>Google</b>                   | TPU v6               | 85                                   | 25                    | 2,125                 | 24                    | 6              | 144                 | HBM3e          | Hynix/Samsung        | 306,000                 |
| <b>AWS</b>                      | Trainium 2           | 40                                   | 16                    | 640                   | 24                    | 4              | 96                  | HBM3e          | Samsung              | 61,440                  |
|                                 | Trainium 2.5         | 30                                   | 16                    | 480                   | 36                    | 4              | 144                 | HBM3e 12hi     | Samsung              | 69,120                  |
|                                 | Trainium 3           | 18                                   | 10                    | 180                   | 36                    | 8              | 288                 | HBM3e 12hi     | Hynix/Samsung        | 51,840                  |
| <b>Tesla</b>                    | Dojo 1               | NA                                   | NA                    | 200                   | 32                    | 5              | 160                 | HBM2e/HBM3     | Samsung              | 1,280                   |
|                                 | FSD                  | NA                                   | NA                    | 1,600                 | NA                    | NA             | NA                  | NA             | NA                   | -                       |
| <b>Microsoft</b>                | Maia 200             | 2                                    | 29                    | 58                    | 16                    | 4              | 64                  | HBM3           | Samsung              | 3,712                   |
|                                 | Maia 200 Enhancement | 1                                    | 29                    | 29                    | 16                    | 4              | 64                  | HBM3           | Samsung              | 1,856                   |
| <b>Chinese GPU/ASIC</b>         | Chinese GPU/ASIC     | 5.5                                  | 20                    | 110                   | 16                    | 4              | 64                  | NA             | Samsung              | 7,040                   |
| <b>Total</b>                    |                      | <b>709</b>                           |                       |                       |                       |                |                     |                |                      | <b>2,426,132</b>        |
| <b>Total HBM demand (mn Gb)</b> |                      |                                      |                       |                       |                       |                |                     |                |                      | <b>19,409</b>           |

Source: Company data, Morgan Stanley Research (e) estimates



# ASE: Earnings Estimates Revision and Quarterly Financials

We fine-tune our 2024-26 revenue estimates by 0-1% and introduce our 2027 financial forecasts. We now see lower margin expansion ahead for the ATM business vs. our previous forecast, which was a little too optimistic on margin expansion in both GM and OPM. Right now we still expect COGS and opex to decrease annually in terms of the percentage of total revenue, but with the margin downward revision this time, our 2025/26 EPS fall by 11-13%. To reflect the opportunity in FAU packaging and testing, we introduce our 2027 forecasts with revenue up 16% y/y and overall margin up 0.5 ppts y/y.

**Exhibit 18:** ASEH: Estimate revisions summary

| (NTS mn)         | New 2024e | Old 2024e | Diff. % | New 2025e | Old 2025e | Diff. % | New 2026e | Old 2026e | Diff. % | New 2027e | Old 2027e | Diff. % |
|------------------|-----------|-----------|---------|-----------|-----------|---------|-----------|-----------|---------|-----------|-----------|---------|
| Net sales        | 595,410   | 588,805   | 1%      | 649,477   | 644,441   | 1%      | 746,299   | 739,484   | 1%      | 864,403   |           |         |
| IC ATM           | 324,908   | 323,502   | 0%      | 372,588   | 369,125   | 1%      | 436,510   | 431,702   | 1%      | 528,925   |           |         |
| EMS              | 271,899   | 270,253   | 1%      | 289,703   | 287,957   | 1%      | 325,194   | 322,946   | 1%      | 354,664   |           |         |
| Gross profit     | 97,708    | 97,271    | 0%      | 121,433   | 128,710   | -6%     | 142,341   | 153,533   | -7%     | 169,793   |           |         |
| Operating profit | 40,830    | 40,663    | 0%      | 61,249    | 68,619    | -11%    | 78,199    | 89,823    | -13%    | 97,905    |           |         |
| Pretax income    | 42,279    | 42,147    | 0%      | 60,292    | 67,762    | -11%    | 77,239    | 88,964    | -13%    | 96,946    |           |         |
| Net income       | 32,286    | 32,166    | 0%      | 45,472    | 51,082    | -11%    | 58,283    | 67,102    | -13%    | 73,146    |           |         |
| Diluted EPS      | 7.37      | 7.34      | 0%      | 10.35     | 11.63     | -11%    | 13.27     | 15.28     | -13%    | 16.66     |           |         |
| <b>Margins</b>   |           |           |         |           |           |         |           |           |         |           |           |         |
| Gross margin     | 16.4%     | 16.5%     |         | 18.7%     | 20.0%     |         | 19.1%     | 20.8%     |         | 19.6%     |           |         |
| IC ATM GM        | 22.4%     | 22.4%     |         | 25.4%     | 27.6%     |         | 25.5%     | 28.4%     |         | 25.8%     |           |         |
| Operating margin | 6.9%      | 6.9%      |         | 9.4%      | 10.6%     |         | 10.5%     | 12.1%     |         | 11.3%     |           |         |
| Pretax margin    | 7.1%      | 7.2%      |         | 9.3%      | 10.5%     |         | 10.3%     | 12.0%     |         | 11.2%     |           |         |
| Net margin       | 5.4%      | 5.5%      |         | 7.0%      | 7.9%      |         | 7.8%      | 9.1%      |         | 8.5%      |           |         |

Source: Company data, Morgan Stanley Research (e) estimates

**Exhibit 19:** ASEH: Quarterly financials

| (NTS mn)                   | 1Q24      | 2Q24      | 3Q24      | 4Q24e     | 1Q25e     | 2Q25e     | 3Q25e     | 4Q25e     | 2022      | 2023      | 2024e     | 2025e     | 2026e     | 2027e     |
|----------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Total Revenues             | 132,803   | 140,238   | 160,105   | 162,254   | 141,163   | 148,594   | 178,639   | 181,082   | 670,873   | 581,914   | 595,410   | 649,477   | 746,299   | 864,403   |
| QOQ %                      | -17.3%    | 5.6%      | 14.2%     | 1.3%      | -13.0%    | 5.3%      | 20.2%     | 1.4%      |           |           |           |           |           |           |
| Y/Y %                      | 1.5%      | 2.9%      | 3.9%      | 1.0%      | 6.3%      | 6.0%      | 11.6%     | 11.6%     | 17.7%     | -13.3%    | 2.3%      | 8.1%      | 14.9%     | 15.8%     |
| Cost of Sales              | (111,935) | (117,172) | (133,732) | (134,863) | (115,836) | (120,954) | (144,713) | (146,541) | (535,943) | (490,157) | (497,702) | (528,044) | (603,958) | (684,610) |
| Percent of Revenues        | 84.3%     | 83.6%     | 83.5%     | 83.1%     | 82.1%     | 81.4%     | 81.0%     | 80.9%     | 79.9%     | 84.2%     | 83.6%     | 81.3%     | 80.9%     | 80.4%     |
| Gross Profit               | 20,868    | 23,066    | 26,373    | 27,401    | 25,327    | 27,640    | 33,926    | 34,540    | 134,930   | 91,757    | 97,708    | 121,433   | 142,341   | 169,793   |
| Gross Margin %             | 15.7%     | 16.4%     | 16.5%     | 16.9%     | 17.9%     | 18.6%     | 19.0%     | 19.1%     | 20.1%     | 15.8%     | 16.4%     | 18.7%     | 19.1%     | 19.6%     |
| Total Operating Expense    | (13,343)  | (14,045)  | (14,957)  | (14,533)  | (13,779)  | (14,491)  | (15,583)  | (16,332)  | (64,754)  | (51,429)  | (56,878)  | (60,185)  | (64,142)  | (71,888)  |
| Percent of Revenues        | 10.0%     | 10.0%     | 9.3%      | 9.0%      | 9.8%      | 9.8%      | 8.7%      | 9.0%      | 9.6%      | 8.8%      | 9.6%      | 9.3%      | 8.6%      | 8.3%      |
| R&D                        | (6,608)   | (7,108)   | (7,439)   | (7,850)   | (7,209)   | (7,705)   | (7,639)   | (8,750)   | (24,370)  | (25,499)  | (28,105)  | (31,305)  | (32,955)  | (38,555)  |
| Percent of Revenues        | 5.0%      | 5.1%      | 4.6%      | 4.9%      | 5.1%      | 5.2%      | 4.3%      | 4.8%      | 3.6%      | 4.4%      | 4.9%      | 4.8%      | 4.4%      | 4.2%      |
| Sales and Marketing        | (1,752)   | (1,938)   | (1,936)   | (1,702)   | (1,770)   | (1,904)   | (2,144)   | (2,173)   | (6,921)   | (8,569)   | (7,327)   | (7,991)   | (8,042)   | (9,319)   |
| % Revenues                 | 1.3%      | 1.4%      | 1.2%      | 1.0%      | 1.3%      | 1.3%      | 1.2%      | 1.2%      | 1.0%      | 1.1%      | 1.2%      | 1.2%      | 1.1%      | 1.1%      |
| General and Admin          | (4,982)   | (5,002)   | (5,582)   | (4,880)   | (4,800)   | (4,881)   | (5,800)   | (5,409)   | (23,464)  | (19,361)  | (20,446)  | (20,889)  | (23,146)  | (26,014)  |
| % Revenues                 | 3.8%      | 3.6%      | 3.5%      | 3.0%      | 3.4%      | 3.3%      | 3.2%      | 3.0%      | 3.5%      | 3.3%      | 3.4%      | 3.2%      | 3.1%      | 3.0%      |
| Operating Income           | 7,525     | 9,021     | 11,416    | 12,868    | 11,548    | 13,149    | 18,343    | 18,208    | 80,176    | 40,328    | 40,830    | 61,249    | 78,199    | 97,905    |
| Operating Margin           | 5.7%      | 6.4%      | 7.1%      | 7.9%      | 8.2%      | 8.9%      | 10.3%     | 10.1%     | 12.0%     | 6.9%      | 6.9%      | 9.4%      | 10.5%     | 11.3%     |
| Total Non-operating Income | 335       | 1,097     | 784       | (767)     | (233)     | (247)     | (237)     | (239)     | 1,450     | 2,273     | 1,449     | (957)     | (960)     | (959)     |
| Profit Before Taxes        | 7,860     | 10,117    | 12,200    | 12,102    | 11,315    | 12,902    | 18,105    | 17,969    | 81,626    | 42,600    | 42,279    | 60,292    | 77,239    | 96,946    |
| Percent of Revenues        | 5.9%      | 7.2%      | 7.6%      | 7.5%      | 8.0%      | 8.7%      | 10.1%     | 9.9%      | 12.2%     | 7.3%      | 7.1%      | 9.3%      | 10.3%     | 11.2%     |
| Taxes                      | (1,904)   | (1,952)   | (2,040)   | (2,420)   | (2,520)   | (2,645)   | (3,712)   | (3,684)   | (16,399)  | (9,043)   | (8,317)   | (12,360)  | (15,834)  | (19,874)  |
| Tax Rate                   | 24.2%     | 19.3%     | 16.7%     | 20.0%     | 20.5%     | 20.5%     | 20.5%     | 20.5%     | 20.1%     | 21.2%     | 19.7%     | 20.5%     | 20.5%     | 20.5%     |
| Net Income, Cont Ops       | 5,955     | 8,165     | 10,160    | 9,681     | 8,796     | 10,257    | 14,394    | 14,286    | 65,227    | 33,557    | 33,961    | 47,932    | 61,405    | 77,072    |
| Minority Interest          | (274)     | (382)     | (521)     | (499)     | (462)     | (535)     | (744)     | (719)     | (3,137)   | (1,832)   | (1,676)   | (2,460)   | (3,122)   | (3,926)   |
| Reported Income (TW GAAP)  | 5,682     | 7,783     | 9,639     | 9,182     | 8,334     | 9,722     | 13,650    | 13,567    | 62,090    | 31,725    | 32,286    | 45,472    | 58,283    | 73,146    |
| Net margin                 | 4.3%      | 5.6%      | 6.0%      | 5.5%      | 6.0%      | 6.5%      | 7.6%      | 7.5%      | 9.3%      | 5.5%      | 5.4%      | 7.0%      | 7.8%      | 8.5%      |
| Y/Y %                      | -2.3%     | 0.6%      | 8.6%      | -2.3%     | 50.2%     | 24.9%     | 41.6%     | 47.8%     | -2.8%     | -46.9%    | 1.6%      | 40.6%     | 28.2%     | 25.5%     |
| EPS for Consensus (NTS)    | 1.30      | 1.78      | 2.19      | 2.09      | 1.94      | 2.21      | 3.11      | 3.08      | 14.35     | 7.30      | 7.37      | 10.35     | 13.27     | 16.66     |
| Seq %                      | -39.7%    | 36.5%     | 23.6%     | -4.7%     | -7.1%     | 13.9%     | 40.4%     | -0.6%     | -2.1%     | -49.1%    | 0.9%      | 40.6%     | 28.2%     | 25.5%     |
| Reported basic EPS (NTS)   | 1.32      | 1.80      | 2.23      | 2.12      | 1.97      | 2.25      | 3.16      | 3.14      | 14.42     | 7.39      | 7.48      | 10.52     | 13.49     | 16.93     |

Source: Company data, Morgan Stanley Research (e) estimates

## ASE: Valuation methodology

We roll-over our valuation to 2025 and raise our price target from NT\$180 to NT\$198. We continue to derive our price target using the base case value from a residual income model and roll-over our valuation to 2025 as we introduce our 2027 financial forecasts. All of our key assumptions are unchanged, including a cost of equity of 9.8% (beta of 1.3, risk premium of 6%, and risk-free rate of 2%), an intermediate growth rate of 7.0%, and a terminal growth rate of 3.0%. Our bull case and bear case targets are also raised to NT\$130 and NT\$240 accordingly.

**Exhibit 20:** ASEH: Residual Income model

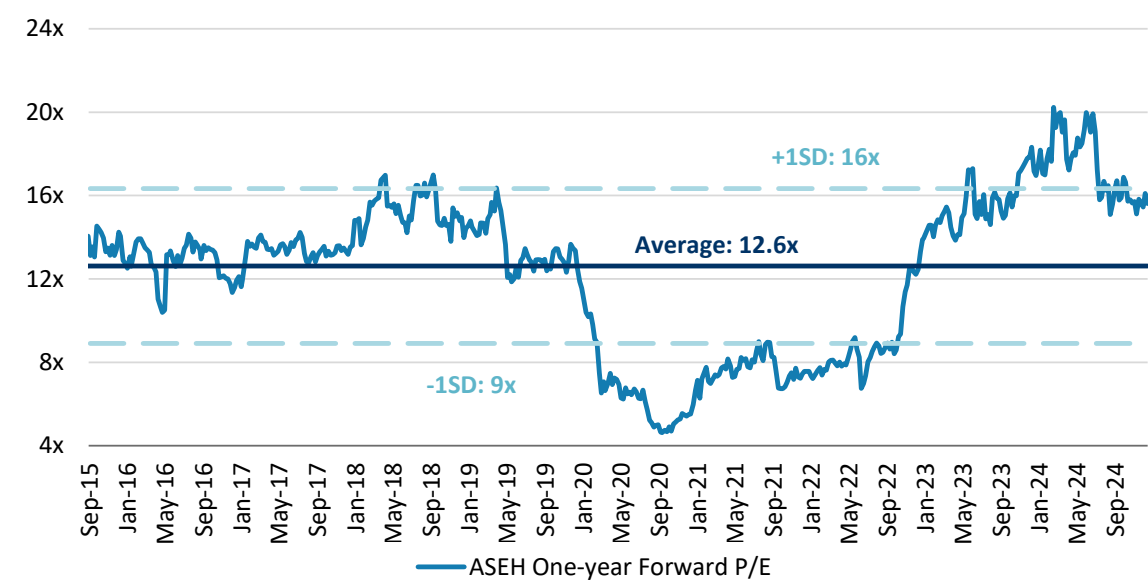
|                        | 2025e          | 2026e   | 2027e   | 2028e   | 2029e   | 2030e   | 2031e   | 2032e   | 2033e   | 2034e   | 2035e   | 2036e   |
|------------------------|----------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| <b>Total Equity</b>    | 357,461        | 388,461 | 426,637 | 462,639 | 501,162 | 542,381 | 586,486 | 633,678 | 684,173 | 738,203 | 796,015 | 857,874 |
| <b>Net Profit</b>      | 45,472         | 58,283  | 73,146  | 78,266  | 83,745  | 89,607  | 95,880  | 102,591 | 109,772 | 117,457 | 125,679 | 134,476 |
| <b>ROAE</b>            | 13.2%          | 15.6%   | 17.9%   | 17.6%   | 17.4%   | 17.2%   | 17.0%   | 16.8%   | 16.7%   | 16.5%   | 16.4%   | 16.3%   |
| <b>Residual Income</b> | 11,276         | 20,830  | 31,651  | 33,287  | 35,059  | 36,954  | 38,980  | 41,148  | 43,466  | 45,946  | 48,599  | 51,437  |
| <b>Spread</b>          | 3.4%           | 5.8%    | 8.1%    | 7.8%    | 7.6%    | 7.4%    | 7.2%    | 7.0%    | 6.9%    | 6.7%    | 6.6%    | 6.5%    |
| Ending Equity Capital  | 357,461        |         |         |         |         |         |         |         |         |         |         |         |
| PV of Forecast Period  | 220,739        |         |         |         |         |         |         |         |         |         |         |         |
| PV of Continuing Value | 278,597        |         |         |         |         |         |         |         |         |         |         |         |
| <b>Equity Value</b>    | <b>856,798</b> |         |         |         |         |         |         |         |         |         |         |         |
| No. of Shares          | 4,322          |         |         |         |         |         |         |         |         |         |         |         |
| <b>Projected Price</b> | <b>198</b>     |         |         |         |         |         |         |         |         |         |         |         |

Source: Company data, Morgan Stanley Research (e) estimates

## Relative valuation

Our price target implies 19x our new 2025 EPS estimate, higher than +1 STD from its historical average of 16x since 2015. The stock is now trading at 16x our 2025 EPS estimate, which we view as attractive amid this AI proliferation and CPO trend.

Exhibit 21: ASEH: One-year forward P/E chart



Source: Company data, Morgan Stanley Research (e) estimates

## Risk Reward – ASE Technology Holding Co. Ltd. (3711.TW)

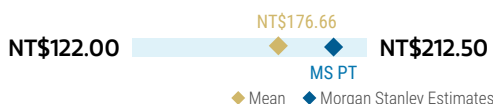
Defensive play with AI growth opportunity; OW

### PRICE TARGET NT\$198.00

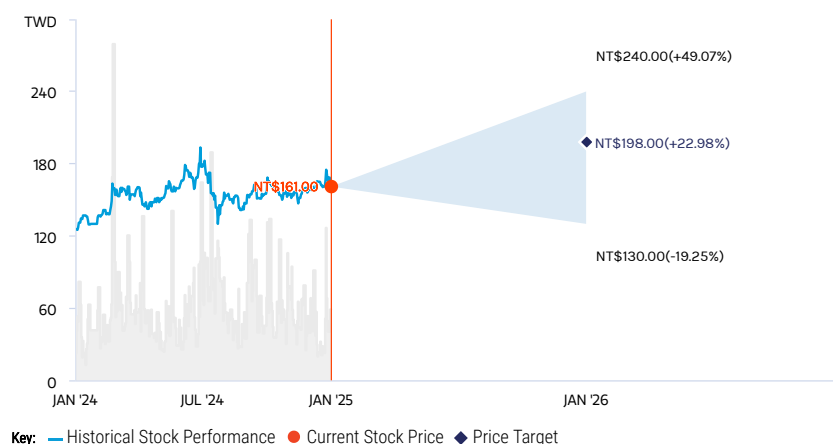
Base case, residual income model. Key assumptions: cost of equity of 9.8%, intermediate growth rate of 7.0%, terminal growth rate of 3.0%, and cash dividend payout ratio of 60%.

#### Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



### RISK REWARD CHART

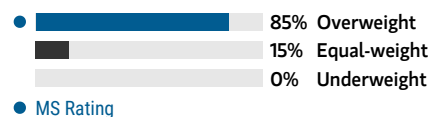


Source: Refinitiv, Morgan Stanley Research

### OVERWEIGHT THESIS

- We think ASE's valuation looks attractive with limited downside risk to our 2025 and 2026 earnings forecasts.
- Secular growth of AI chip packaging and testing should drive a re-rating to our price target of NT\$198.
- Our price target implies 16x our 2025e EPS, higher than ASE's 10-year average P/E multiple of 12x.
- We think the potential of its advanced packaging business is underestimated. TSMC's CoWoS outsourcing story is still convincing.
- ASE will be one of the beneficiaries amid CPO trend.

#### Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

#### Risk Reward Themes

Secular Growth: *Positive*  
Technology Diffusion: *Positive*

View descriptions of Risk Rewards Themes [here](#)

### BULL CASE

NT\$240.00

#### 20x 2025e EPS

**Strong semi growth; competition becomes rational thanks to merger; upside for SiP penetration rate with mainstream products:** Competition becomes rational with sustained gross margin. Stronger price bargaining power and technology advancement thanks to smooth consolidation between ASE and SPIL. SiP shipments accelerate among mmWave AiP, camera modules, acoustic modules, 3D sensing, and wearables. ASEH develops comprehensive advanced packaging technology with high traction among customers.

### BASE CASE

NT\$198.00

#### 16x 2025e EPS

**Cycle bottoming with AI/advanced packaging acting as the new driving force:** Demand for wire-bonding and flip-chip from consumer tech demand remains weak, and we only see a weak recovery approaching. However, the advanced packaging business should be the major source of margin accretion in coming years. Pricing for ATM businesses remains more defensive than peers. CoWoS outsourcing from TSMC may aid its top line and margin profile as well as CPO opportunity in 2026.

### BEAR CASE

NT\$130.00

#### 11x 2025e EPS

**Semis enter down-cycle, with less back-end outsourcing and big customers searching for more cost-effective alternatives:** The macro situation weakens, impairing semi demand. The need for second sources causes big customers to shift back-end orders away from ASE, along with pricing erosion given competition. SiP adoption remains limited outside of wearables. Gross margins keep deteriorating given competition from Chinese peers.

## Risk Reward – ASE Technology Holding Co. Ltd. (3711.TW)

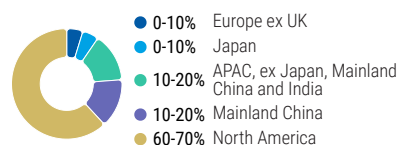
## KEY EARNINGS INPUTS

| Drivers                                         | 2023    | 2024e   | 2025e   | 2026e   |
|-------------------------------------------------|---------|---------|---------|---------|
| Revenue from IC packaging Segment (NT\$, mn)    | 260,486 | 266,212 | 302,774 | 353,903 |
| Revenue from Testing Segment (NT\$, mn)         | 49,881  | 53,255  | 64,306  | 76,434  |
| Revenue from Direct Material Segment (NT\$, mn) | 4,571   | 5,192   | 5,508   | 6,172   |
| Revenue from SiP (NT\$, mn)                     | 87,170  | 98,302  | 108,615 | 119,268 |

## INVESTMENT DRIVERS

- Global semi growth
- New applications for SiP
- Synergies from merger with SPIL

## GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate  
View explanation of regional hierarchies [here](#)

## MS ALPHA MODELS

**3/5**  
**MOST** 3 Month  
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

## RISKS TO PT/RATING

## RISKS TO UPSIDE

- Stronger-than-expected global economic and semiconductor growth
- Margin expansion through merger consolidation
- New wearables project wins with higher margins

## RISKS TO DOWNSIDE

- Slower-than-expected global economic and semiconductor growth
- Further margin erosion amid price competition
- Lack of new features in wearables, causing consumers to lose interest

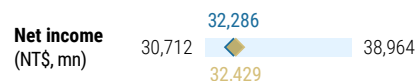
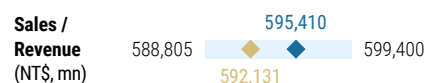
## OWNERSHIP POSITIONING

Inst. Owners, % Active 71.1%

Source: Refinitiv, Morgan Stanley Research

## MS ESTIMATES VS. CONSENSUS

FY Dec 2024e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

## ASE: Financial Summary

## Exhibit 22: ASE: Financial Summary

## Income Statements

| NT\$mn (Years End Dec)          | 2023          | 2024e         | 2025e         | 2026e         | 2027e         |
|---------------------------------|---------------|---------------|---------------|---------------|---------------|
| Net sales                       | 581,914       | 595,410       | 649,477       | 740,299       | 864,402       |
| COGS                            | (490,157)     | (497,702)     | (528,044)     | (603,958)     | (694,610)     |
| Depreciation Expense            | (52,135)      | (55,940)      | (61,236)      | (61,236)      | (61,236)      |
| Variable Cost                   | (438,022)     | (441,762)     | (466,808)     | (542,722)     | (633,374)     |
| Gross profit                    | 91,757        | 97,708        | 121,433       | 142,341       | 169,793       |
| Operating expenses              | (51,429)      | (56,878)      | (60,185)      | (64,142)      | (71,888)      |
| <b>Operating income</b>         | <b>40,328</b> | <b>40,830</b> | <b>61,249</b> | <b>78,199</b> | <b>97,905</b> |
| Non-operating income            | 2,273         | 1,449         | (957)         | (960)         | (959)         |
| Pre-tax income                  | 42,600        | 42,279        | 60,292        | 77,239        | 96,946        |
| Income tax                      | (10,875)      | (9,993)       | (14,820)      | (18,956)      | (23,800)      |
| <b>Reported net income</b>      | <b>31,725</b> | <b>32,286</b> | <b>45,472</b> | <b>58,283</b> | <b>73,146</b> |
| Adj. wtd.avg.shrs (m)           | 4,345         | 4,384         | 4,391         | 4,391         | 4,391         |
| <b>Basic EPS (NT\$)</b>         | <b>7.39</b>   | <b>7.48</b>   | <b>10.52</b>  | <b>13.49</b>  | <b>16.93</b>  |
| <b>EPS for Consensus (NT\$)</b> | <b>7.30</b>   | <b>7.37</b>   | <b>10.35</b>  | <b>13.27</b>  | <b>16.66</b>  |

## Balance Sheets

| NT\$mn (Years End Dec)                    | 2023           | 2024e          | 2025e          | 2026e          | 2027e          |
|-------------------------------------------|----------------|----------------|----------------|----------------|----------------|
| Cash                                      | 67,285         | 49,609         | 60,468         | 68,931         | 67,398         |
| Mar Securities                            | 4,085          | 4,085          | 4,085          | 4,085          | 4,085          |
| AR/INR                                    | 99,529         | 107,088        | 109,572        | 119,521        | 137,239        |
| Inventory                                 | 68,375         | 83,466         | 84,751         | 89,918         | 102,844        |
| Other                                     | 22,074         | 22,074         | 22,074         | 22,074         | 22,074         |
| <b>Current Assets</b>                     | <b>261,348</b> | <b>266,322</b> | <b>280,950</b> | <b>304,529</b> | <b>333,741</b> |
| Long-term investments                     | 44,303         | 45,788         | 46,588         | 47,388         | 48,188         |
| Fixed assets                              | 264,812        | 277,086        | 292,381        | 310,565        | 332,537        |
| Other assets                              | 96,113         | 96,113         | 96,113         | 96,113         | 96,113         |
| <b>Total Assets</b>                       | <b>666,575</b> | <b>685,309</b> | <b>716,032</b> | <b>758,595</b> | <b>810,580</b> |
| S/T borrowings                            | 37,737         | 37,737         | 37,737         | 37,737         | 37,737         |
| AP/NP                                     | 70,329         | 75,812         | 80,434         | 91,998         | 105,806        |
| Other ST liabilities                      | 111,495        | 111,495        | 111,495        | 111,495        | 111,495        |
| LT debt                                   | 101,854        | 101,854        | 101,854        | 101,854        | 101,854        |
| Other LT liabilities                      | 24,263         | 24,263         | 24,263         | 24,263         | 24,263         |
| <b>Total Liabilities</b>                  | <b>348,466</b> | <b>353,949</b> | <b>358,571</b> | <b>370,134</b> | <b>383,943</b> |
| Common stocks                             | 43,802         | 43,802         | 43,802         | 43,802         | 43,802         |
| Preferred stocks                          | 0              | 0              | 0              | 0              | 0              |
| Capital Reserve                           | 144,407        | 144,407        | 144,407        | 144,407        | 144,407        |
| Retained earning                          | 131,860        | 145,110        | 171,211        | 202,211        | 240,387        |
| Minority Equity                           | 0              | 0              | 0              | 0              | 0              |
| Others                                    | (1,959)        | (1,959)        | (1,959)        | (1,959)        | (1,959)        |
| <b>Total Equity</b>                       | <b>318,110</b> | <b>331,360</b> | <b>357,461</b> | <b>388,461</b> | <b>426,637</b> |
| <b>Total Liab. &amp; Shrhldr's Equity</b> | <b>666,575</b> | <b>685,309</b> | <b>716,032</b> | <b>758,595</b> | <b>810,580</b> |

e = Morgan Stanley Research Estimates

Source: Morgan Stanley Research, company data

## Cash Flow Statements

| NT\$mn (Years End Dec)          | 2023            | 2024e           | 2025e           | 2026e           | 2027e           |
|---------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| <b>Cashflow from Operations</b> | <b>114,422</b>  | <b>76,243</b>   | <b>114,067</b>  | <b>126,436</b>  | <b>131,666</b>  |
| Net profits                     | 31,725          | 32,286          | 45,472          | 58,283          | 73,146          |
| Depreciation                    | 54,195          | 62,609          | 68,541          | 72,506          | 76,256          |
| Working Capital Change          | 13,818          | (17,167)        | 854             | (3,553)         | (16,936)        |
| Other adjustments               | 14,683          | (1,485)         | (800)           | (800)           | (800)           |
| <b>Cashflow from Investing</b>  | <b>(55,122)</b> | <b>(74,883)</b> | <b>(83,836)</b> | <b>(90,691)</b> | <b>(98,229)</b> |
| Capex                           | (53,683)        | (74,883)        | (83,836)        | (90,691)        | (98,229)        |
| Change of LT Investment         | 323             | 0               | 0               | 0               | 0               |
| Change of ST Investment         | 4               | 0               | 0               | 0               | 0               |
| Other adjustments               | (1,766)         | 0               | 0               | 0               | 0               |
| <b>Cashflow from financing</b>  | <b>(49,101)</b> | <b>(19,035)</b> | <b>(19,372)</b> | <b>(27,283)</b> | <b>(34,970)</b> |
| Increase in S/T debt            | (14,263)        | 0               | 0               | 0               | 0               |
| Increase in S/T debt            | 6,019           | 0               | 0               | 0               | 0               |
| Cash Dividend Paid              | (37,841)        | (19,035)        | (19,372)        | (27,283)        | (34,970)        |
| Dir& Emp Bonus Paid             | 0               | 0               | 0               | 0               | 0               |
| Other adjustments               | (3,016)         | 0               | 0               | 0               | 0               |
| Exchange rate adjustment        | (955)           | 0               | 0               | 0               | 0               |
| <b>Net change in cash</b>       | <b>9,244</b>    | <b>-17,675</b>  | <b>10,859</b>   | <b>8,462</b>    | <b>-1,532</b>   |

## Financial Ratios

|                           | 2023  | 2024e | 2025e | 2026e | 2027e |
|---------------------------|-------|-------|-------|-------|-------|
| <b>Growth(%)</b>          |       |       |       |       |       |
| Turnover                  | -13.3 | 2.3   | 9.1   | 14.9  | 15.8  |
| Operating profits         | -49.7 | 1.2   | 50.0  | 27.7  | 25.2  |
| Pretax profits            | -47.8 | -0.8  | 42.6  | 28.1  | 25.5  |
| Net profits               | -48.9 | 1.8   | 40.8  | 28.2  | 25.5  |
| EPS                       | -49.1 | 0.9   | 40.6  | 28.2  | 25.5  |
| <b>Margins (%)</b>        |       |       |       |       |       |
| Gross Margin              | 15.8  | 16.4  | 18.7  | 19.1  | 19.6  |
| Operating Margin          | 6.9   | 6.9   | 9.4   | 10.5  | 11.3  |
| EBITA Margin              | 15.9  | 16.3  | 18.9  | 18.7  | 18.4  |
| Pretax Margin             | 7.3   | 7.1   | 9.3   | 10.3  | 11.2  |
| Net Profit                | 5.5   | 5.4   | 7.0   | 7.8   | 8.5   |
| <b>Return (%)</b>         |       |       |       |       |       |
| ROAE                      | 9.9   | 9.9   | 13.2  | 15.6  | 17.9  |
| ROAA                      | 4.6   | 4.8   | 6.5   | 7.9   | 9.3   |
| <b>Gearing (%)</b>        |       |       |       |       |       |
| Net Debt/Equity           | (9.3) | (3.6) | (6.4) | (8.0) | (7.0) |
| Liabilities/Equity        | 109.5 | 106.8 | 100.3 | 95.3  | 90.0  |
| <b>Ratios (X)</b>         |       |       |       |       |       |
| Current ratio             | 1.2   | 1.2   | 1.2   | 1.3   | 1.3   |
| Quick ratio               | 0.8   | 0.7   | 0.7   | 0.8   | 0.8   |
| <b>Others</b>             |       |       |       |       |       |
| AR/INR Turnover (days)    | 63    | 67    | 67    | 67    | 67    |
| Inventory Turnover (days) | 48    | 52    | 52    | 52    | 52    |
| AP Turnover (days)        | 56    | 56    | 56    | 56    | 56    |
| Cash Conversion (days)    | 55    | 64    | 64    | 64    | 64    |

Source: Company data, Morgan Stanley Research (e) estimates

## Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options\\_Probabilities\\_Exhibit\\_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR\\_Themes\\_Exhibit\\_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG\\_Exhibit\\_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG\\_Sustainable\\_Solutions\\_External\\_Link.pdf](#)
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| Stock Rating Category | Coverage Universe |            | Investment Banking Clients (IBC) |                |                      | Other Material Investment Services Clients (MISC) |                       |
|-----------------------|-------------------|------------|----------------------------------|----------------|----------------------|---------------------------------------------------|-----------------------|
|                       | Count             | % of Total | Count                            | % of Total IBC | % of Rating Category | Count                                             | % of Total Other MISC |
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| Equal-weight/Hold     | 1703              | 45%        | 369                              | 46%            | 22%                  | 810                                               | 48%                   |
| Not-Rated/Hold        | 5                 | 0%         | 0                                | 0%             | 0%                   | 2                                                 | 0%                    |
| Underweight/Sell      | 596               | 16%        | 74                               | 9%             | 12%                  | 222                                               | 13%                   |
| Total                 | 3,767             |            | 806                              |                |                      | 1703                                              |                       |

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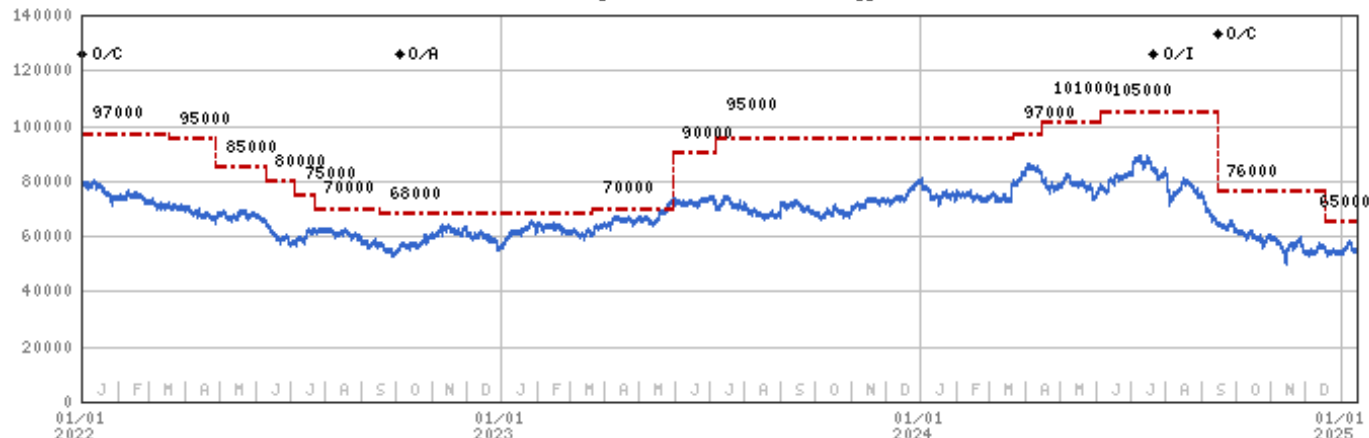
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Samsung Electronics (005930.KS) - As of 01/14/25 GMT in KRW  
Industry : S. Korea Technology



Stock Rating History: 1/1/20 : O/A; 7/19/21 : O/I; 8/12/21 : O/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : O/C

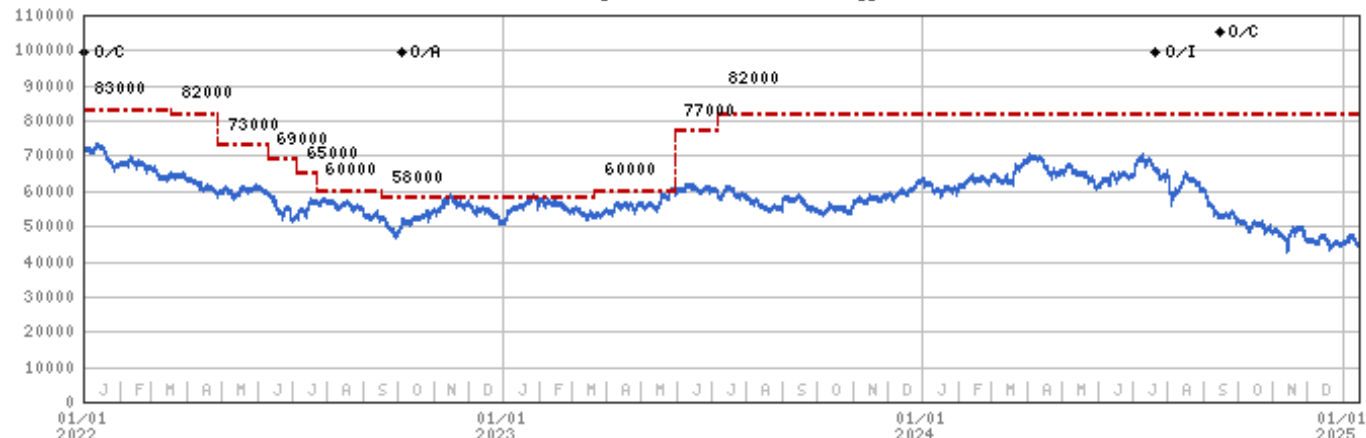
Price Target History: 11/18/19 : 60000; 1/14/20 : 72000; 2/26/20 : 75000; 3/19/20 : 68000; 4/29/20 : 65000; 7/12/20 : 70000;  
9/11/20 : 73000; 11/27/20 : 88000; 1/12/21 : 110000; 2/25/21 : 115000; 5/18/21 : 93000; 6/8/21 : 98000; 8/12/21 : 89000;  
9/15/21 : 95000; 12/3/21 : 97000; 3/18/22 : 95000; 4/28/22 : 85000; 6/10/22 : 80000; 7/5/22 : 75000; 7/22/22 : 70000;  
9/17/22 : 68000; 3/21/23 : 70000; 5/30/23 : 90000; 7/7/23 : 95000; 3/22/24 : 97000; 4/16/24 : 101000; 6/6/24 : 105000;  
9/15/24 : 76000; 12/18/24 : 65000

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Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)  
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Samsung Electronics (005935.KS) - As of 01/14/25 GMT in KRW  
Industry : S. Korea Technology



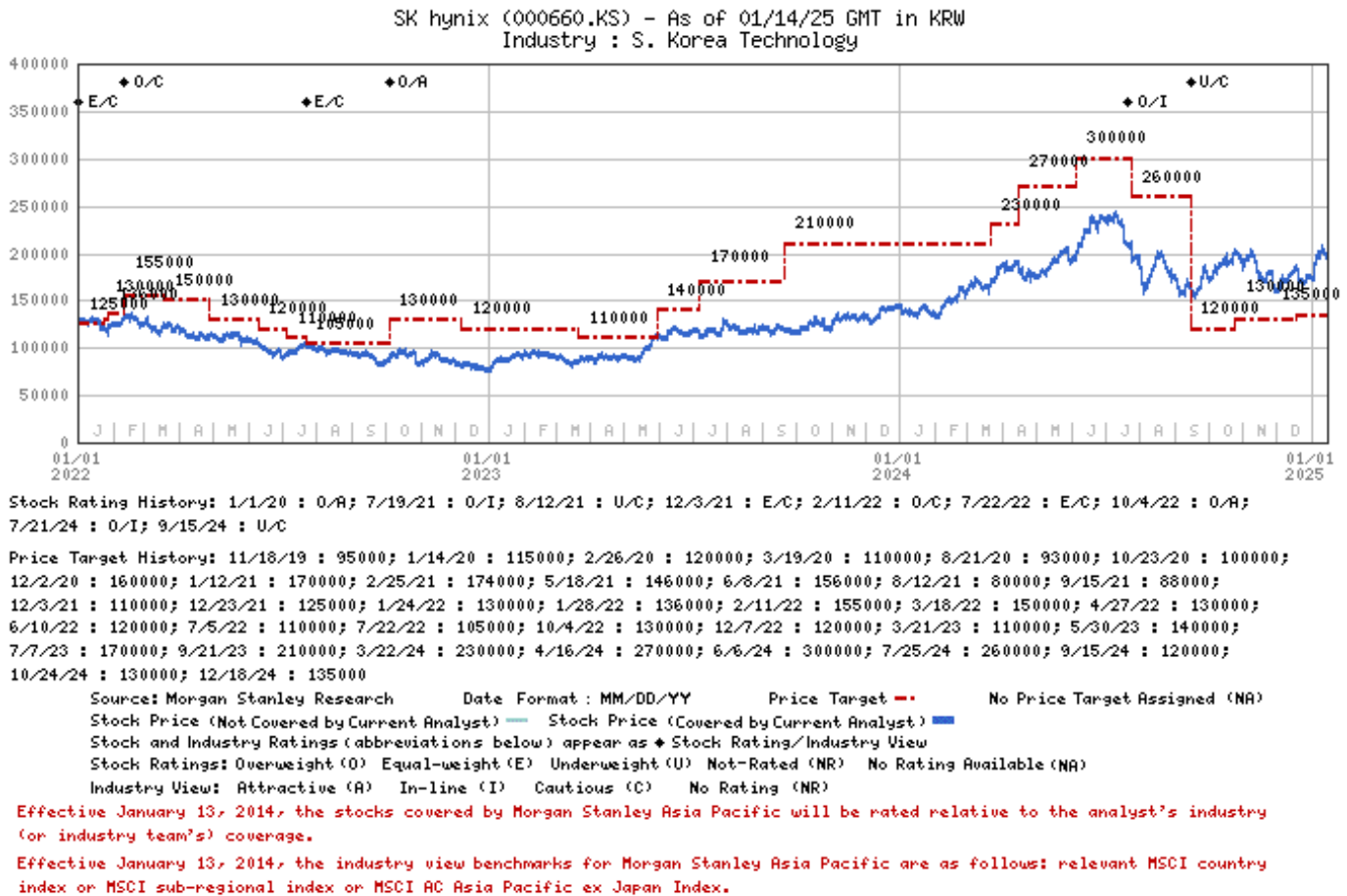
Stock Rating History: 1/1/20 : 0/A; 7/19/21 : 0/I; 8/12/21 : 0/C; 10/4/22 : 0/A; 7/21/24 : 0/I; 9/15/24 : 0/C

Price Target History: 11/18/19 : 49000; 1/14/20 : 59000; 2/26/20 : 62000; 3/19/20 : 56000; 4/29/20 : 53000; 7/12/20 : 57000;  
9/11/20 : 60000; 11/27/20 : 72000; 1/12/21 : 90000; 5/18/21 : 80000; 6/8/21 : 84000; 8/12/21 : 77000; 9/15/21 : 82000;  
12/3/21 : 83000; 3/18/22 : 82000; 4/28/22 : 73000; 6/10/22 : 69000; 7/5/22 : 65000; 7/22/22 : 60000; 9/17/22 : 58000;  
3/21/23 : 60000; 5/30/23 : 77000; 7/7/23 : 82000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)  
Stock Price (Not Covered by Current Analyst) --- Stock Price (Covered by Current Analyst) ---  
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View  
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)  
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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## INDUSTRY COVERAGE: Greater China Technology Semiconductors

| COMPANY (TICKER)                                     | RATING (AS OF) | PRICE* (01/14/2025) |
|------------------------------------------------------|----------------|---------------------|
| <b>Charlie Chan</b>                                  |                |                     |
| ACM Research Inc (ACMR.O)                            | O (03/07/2023) | US\$15.49           |
| Advanced Micro-Fabrication Equipment Inc (688012.SS) | O (11/06/2023) | Rmb188.45           |
| Alchip Technologies Ltd (3661.TW)                    | O (05/14/2021) | NT\$2,845.00        |
| Andes Technology Corp (6533.TW)                      | O (08/04/2022) | NT\$363.00          |
| ASE Technology Holding Co. Ltd. (3711.TW)            | O (09/15/2024) | NT\$161.00          |
| Global Unichip Corp (3443.TW)                        | O (07/27/2024) | NT\$1,205.00        |
| King Yuan Electronics Co Ltd (2449.TW)               | O (03/03/2023) | NT\$111.50          |
| M31 Technology Corp (6643.TWO)                       | E (09/23/2024) | NT\$609.00          |
| MediaTek (2454.TW)                                   | O (09/12/2023) | NT\$1,400.00        |
| Nanya Technology Corp. (2408.TW)                     | U (11/06/2023) | NT\$26.65           |
| Phison Electronics Corp (8299.TWO)                   | E (05/13/2024) | NT\$452.50          |
| Silergy Corp. (6415.TW)                              | E (08/28/2024) | NT\$372.50          |
| SMIC (0981.HK)                                       | U (02/22/2023) | HK\$35.10           |
| TSMC (2330.TW)                                       | O (02/07/2022) | NT\$1,090.00        |
| UMC (2303.TW)                                        | E (10/28/2024) | NT\$42.25           |
| Vanguard International Semiconductor (5347.TWO)      | U (08/05/2024) | NT\$95.20           |
| Will Semiconductor Co Ltd Shanghai (603501.SS)       | E (10/28/2024) | Rmb104.75           |
| <b>Daisy Dai, CFA</b>                                |                |                     |
| ASMP Ltd (0522.HK)                                   | E (11/20/2024) | HK\$74.20           |
| China Resources Microelectronics Limited (688396.SS) | U (09/02/2024) | Rmb46.30            |
| Elan Microelectronics Corp (2458.TW)                 | O (04/29/2024) | NT\$145.00          |
| Empyrean Technology Co Ltd (301269.SZ)               | O (06/26/2023) | Rmb113.14           |
| Hangzhou Silan Microelectronics Co. Ltd. (600460.SS) | E (04/09/2024) | Rmb25.41            |
| JCET Group Co Ltd (600584.SS)                        | U (09/25/2024) | Rmb40.60            |
| Shanghai Anlogic Infotech Co Ltd (688107.SS)         | E (05/14/2024) | Rmb26.76            |
| Shanghai Fudan Microelectronics (1385.HK)            | E (10/18/2022) | HK\$14.20           |

|                                                     |                |              |
|-----------------------------------------------------|----------------|--------------|
| Unigroup Guoxin Microelectronics Co Ltd (002049.SZ) | U (01/10/2023) | Rmb59.87     |
| Universal Scientific Ind. (Shanghai) (601231.SS)    | E (10/23/2024) | Rmb14.85     |
| Yangjie Technology (300373.SZ)                      | O (06/10/2022) | Rmb42.80     |
| <b>Daniel Yen, CFA</b>                              |                |              |
| AP Memory Technology Corp (6531.TW)                 | E (09/15/2024) | NT\$280.00   |
| ASMedia Technology Inc (5269.TW)                    | O (01/23/2024) | NT\$2,045.00 |
| Aspeed Technology (5274.TWO)                        | E (09/15/2024) | NT\$3,290.00 |
| Egis Technology Inc (6462.TWO)                      | E (03/26/2024) | NT\$152.50   |
| Espressif Systems (688018.SS)                       | O (05/15/2023) | Rmb229.00    |
| GigaDevice Semiconductor Beijing Inc (603986.SS)    | U (09/15/2024) | Rmb129.64    |
| Macronix International Co Ltd (2337.TW)             | U (10/19/2021) | NT\$18.25    |
| Montage Technology Co Ltd (688008.SS)               | O (11/08/2023) | Rmb66.43     |
| Novatek (3034.TW)                                   | O (03/18/2024) | NT\$504.00   |
| Nuvoton Technology Corporation (4919.TW)            | O (10/29/2024) | NT\$88.60    |
| Parade Technologies Ltd (4966.TWO)                  | U (06/04/2024) | NT\$688.00   |
| Realtek Semiconductor (2379.TW)                     | U (09/15/2024) | NT\$519.00   |
| Shenzhen Goodix Technology Co Ltd (603160.SS)       | E (06/12/2024) | Rmb77.84     |
| Sino Wealth Electronic (300327.SZ)                  | E (03/18/2024) | Rmb23.39     |
| Winbond Electronics Corp (2344.TW)                  | U (09/15/2024) | NT\$13.55    |
| WPG Holdings (3702.TW)                              | E (11/16/2023) | NT\$69.70    |
| <b>Duan Liu</b>                                     |                |              |
| Dosilicon Co Ltd (688110.SS)                        | U (09/06/2024) | Rmb25.99     |
| Shenzhen Longsys Electronics Co Ltd (301308.SZ)     | U (10/30/2024) | Rmb82.44     |
| <b>Ray Wu, CFA</b>                                  |                |              |
| Advanced Wireless Semiconductor Co (8086.TWO)       | E (10/24/2024) | NT\$100.50   |
| AllRing Tech Co. (6187.TWO)                         | O (08/27/2024) | NT\$367.50   |
| Episil Technologies Inc (3707.TWO)                  | O (06/05/2023) | NT\$43.50    |
| GlobalWafers Co Ltd (6488.TWO)                      | E (02/22/2023) | NT\$338.00   |
| Gudeng Precision (3680.TWO)                         | O (08/27/2024) | NT\$470.00   |
| Hua Hong Semiconductor Ltd (1347.HK)                | O (06/13/2024) | HK\$22.35    |
| Maxscend Microelectronics Co Ltd (300782.SZ)        | U (01/11/2021) | Rmb80.58     |
| NAURA Technology Group Co Ltd (002371.SZ)           | O (11/06/2023) | Rmb391.35    |
| Powerchip Semiconductor Manufacturing Co (6770.TW)  | E (06/13/2024) | NT\$14.10    |
| RichWave Technology Corp. (4968.TW)                 | O (07/05/2024) | NT\$175.50   |
| SG Micro Corp. (300661.SZ)                          | O (06/17/2024) | Rmb76.20     |
| SICC Co Ltd (688234.SS)                             | O (09/18/2023) | Rmb55.75     |
| Silicon Motion (SIMO.O)                             | O (05/06/2024) | US\$51.90    |
| StarPower Semiconductor Ltd (603290.SS)             | O (03/01/2022) | Rmb84.57     |
| UPI Semiconductor Corp. (6719.TW)                   | E (04/18/2023) | NT\$233.00   |
| WIN Semiconductors Corp (3105.TWO)                  | O (06/28/2024) | NT\$96.90    |

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\* Historical prices are not split adjusted.